RELATIONAL MODEL

Definition

- It was introduced by C. F. Codd in 1970.
- It is the most widely used data model.
- It describes the world as “a collection of inter-related relations (or tables).”
- It involves the use of data tables that collect groups of elements into relations.
- These models work based on the idea that each table setup will include a primary key or identifier. Other tables use that identifier to provide "relational" data links and results.

Definition

- Store the data in a simple data structure (tables)
- Access it through a high level set-at-a-time Data Manipulation Language (DML)
- Be independent from physical storage

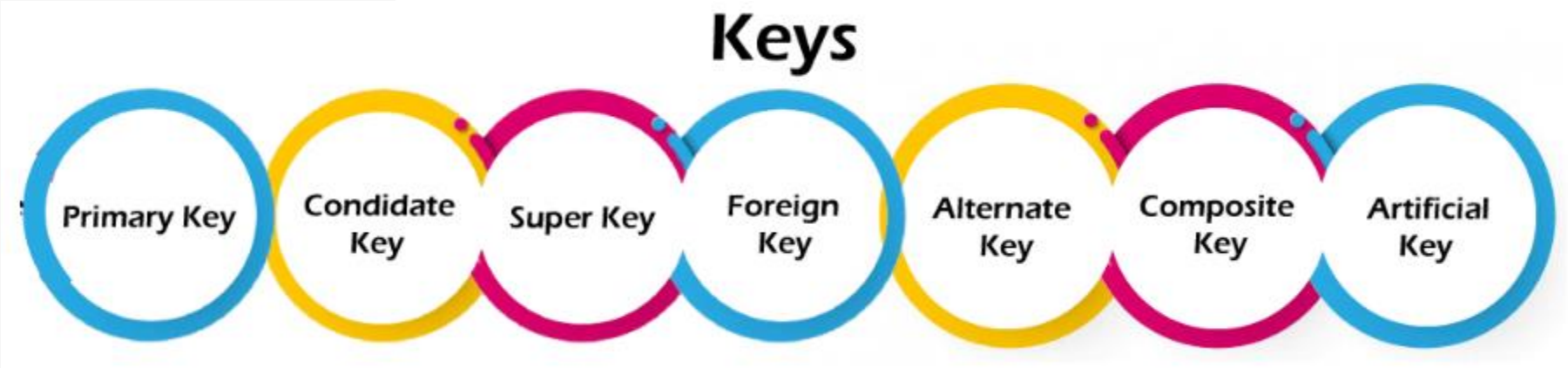
Concepts

- Tables - relations are saved in the format of tables. This format stores the relation among entities. A table has rows and columns, where rows represents records and columns represent the attributes.
- Tuple - a single row of a table, which contains a single record for that relation.
- Relation instance - A finite set of tuples in the relational database system. Relation instances do not have duplicate tuples.

Concepts

- Relation schema - Describes the relation name (table name), attributes, and their names.
- Relation key - Each row has one or more attributes, known as relation key, which can identify the row in the relation (table) uniquely.
- Attribute domain - Every attribute has some pre-defined value scope, known as attribute domain.

Keys


Keys
Primary Key
Condidate Key
Super Key
Foreign Key
Alternate Key
Composite Key
Artificial Key

Candidate Key

- A super key with no repeated attributes.
  - The Primary key should be selected from the candidate keys.
- Every table must have at least a single candidate key.
- A table can have multiple candidate keys but only a single primary key.
- Properties
  - It must contain unique values
  - Must not contain null values
  - It should contain minimum fields to ensure uniqueness
  - Uniquely identify each record in a table

Candidate Key

- Example:
- [Order ID]
- [Customer ID]
- [Customer Contact ID]

Order ID	Order Date	Customer ID	Customer Contact ID	Product ID	Product Name	Quantity
CA-115	6/9/2018	BH-11710	435	10283	Newell 322	3
CA-330	12/9/2020	KB-16585	695	10156	Xerox 232	10
US-630	9/17/2019	TB-21520	214	10168	Lumber Crayons	17
CA-727	10/19/2021	MA-17560	209	10249	Color Paper	8
CA-590	12/8/2020	GH-14485	147	10497	Avery 485	5

Order Table

Primary Key

- A column that uniquely identify every row in that table.
- A table cannot have more than one primary key.
- Rules for defining primary key
  - Two rows can’t have the same primary key value
  - It must for every row to have a primary key value.
  - The primary key field cannot be null.
  - The value in a primary key column can never be modified or updated if any foreign key refers to that primary key.

Primary Key

- Example: Any one of the following candidate keys can be chosen as a Primary key.
  - [Order ID]
  - [Customer ID]
  - [Customer Contact ID]

Order ID	Order Date	Customer ID	Customer Contact ID	Product ID	Product Name	Quantity
CA-115	6/9/2018	BH-11710	435	10283	Newell 322	3
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US-630	9/17/2019	TB-21520	214	10168	Lumber Crayons	17
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Order Table

Super Key

- A group of single or multiple keys which identifies rows in a table.
- May have additional attributes that are not needed for unique identification.
- Example :

Order ID	Order Date	Customer ID	Customer Contact ID	Product ID	Product Name	Quantity
CA-115	6/9/2018	BH-11710	435	10283	Newell 322	3
CA-330	12/9/2020	KB-16585	695	10156	Xerox 232	10
US-630	9/17/2019	TB-21520	214	10168	Lumber Crayons	17
CA-727	10/19/2021	MA-17560	209	10249	Color Paper	8
CA-590	12/8/2020	GH-14485	147	10497	Avery 485	5

Order Table

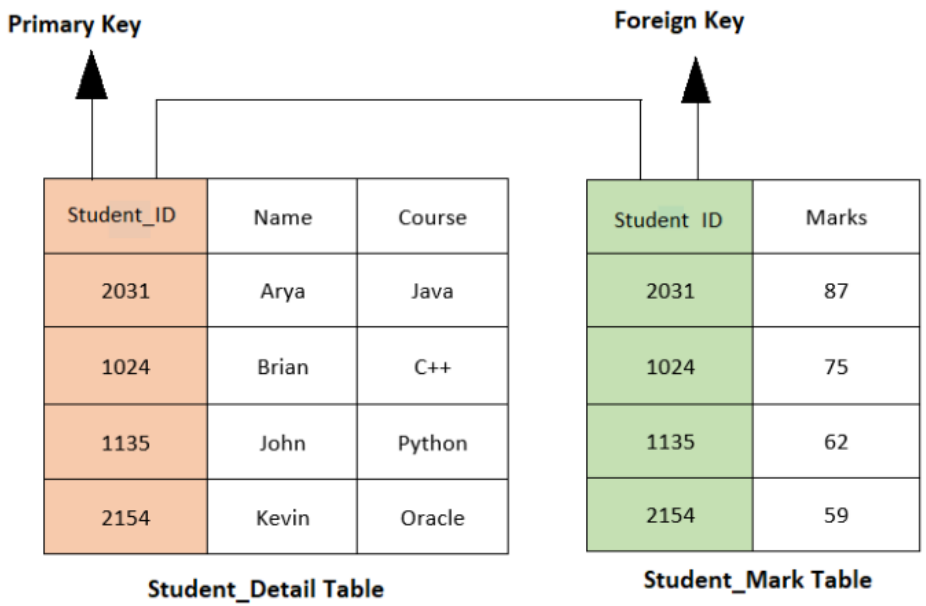
- [Order ID]
- [Customer ID]
- [Customer Contact ID]
- [Order ID, Customer ID]
- [Customer Contact ID, Order ID]
- [Customer ID, Customer Contact ID]
- [Order ID, Customer ID, Customer Contact ID]

Alternate Key

- A column or group of columns in a table that uniquely identify every row in that table.
- A table can have multiple choices for a primary key but only one can be set as the primary key.
- All the candidate keys which are not selected as primary key are called an Alternate Key.
- Example:
  - If the [Order ID] is selected as the Primary Key, then the [Customer ID]and the [Customer Contact ID] can be used as alternate keys.

Foreign Key

- A column that creates a relationship between two tables.
- The purpose of Foreign keys is to maintain data integrity and allow navigation between two different instances of an entity.
- It acts as a cross-reference between two tables as it references the prima



Composite Key

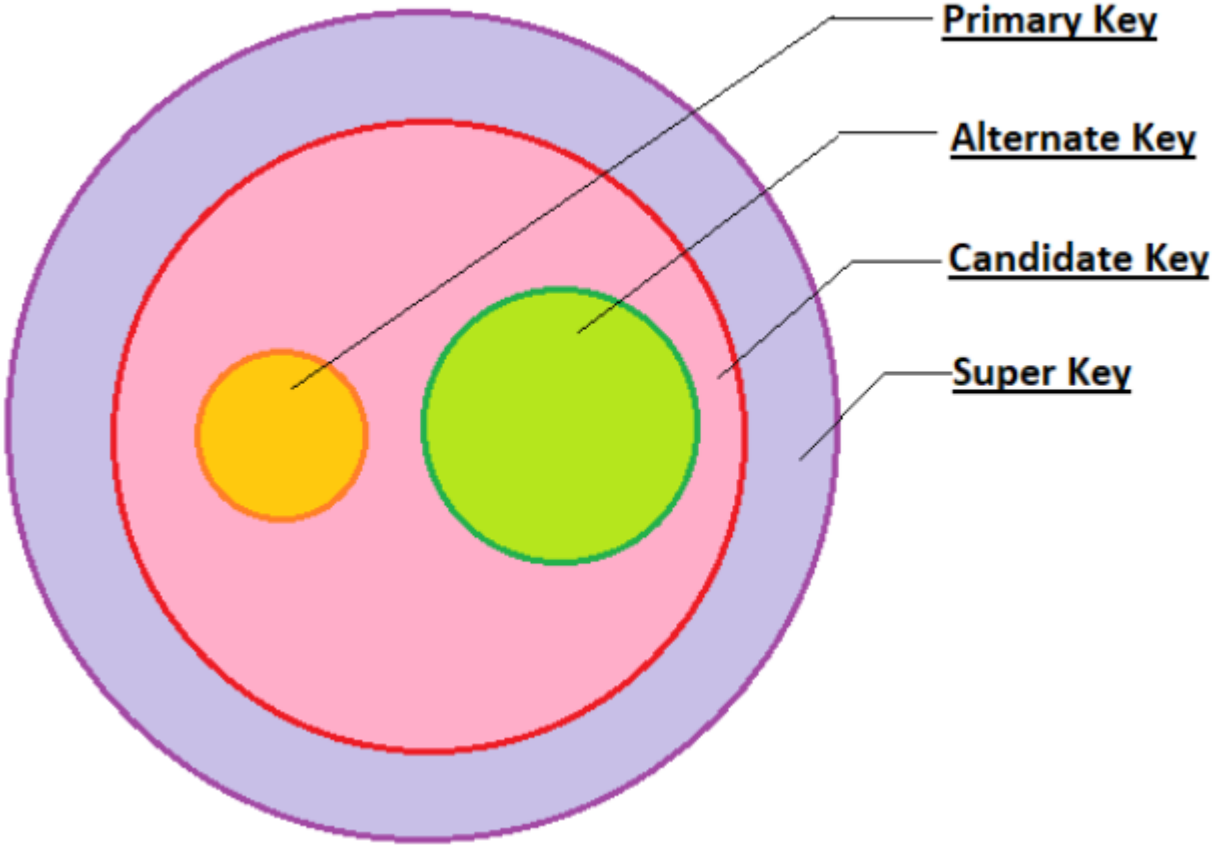
- A combination of two or more columns that uniquely identify rows in a table.
- They are combined to uniquely identify records in a table.

Surrogate Key

- An artificial key which aims to uniquely identify each record.
- This kind of partial key in dbms is unique because it is created when you don't have any natural primary key.
- They do not lend any meaning to the data in the table.
- Surrogate key in DBMS is usually an integer.
- Surrogate Keys of DBMS are allowed in certain cases when:
  - The primary key is too big
  - Complicated Primary Key
  - Absence of key

SR_No.	Name	Age	Gender	State
1	Kathy Will	22	Male	CA
2	Ken Black	27	Female	NY
3	Tracy Blumstein	32	Female	GA
4	Matt Adelman	21	Male	IL
5	Gene Hale	29	Male	AZ

Keys



Constraints

Relational Integrity Constraints :

- Key constraints
- Domain constraints
- Referential integrity constraints

Key Constraints

- There must be at least one minimal subset of attributes in the relation, which can identify a tuple uniquely.
- This minimal subset of attributes is called key for that relation.
- If there are more than one such minimal subsets, these are called candidate keys.

Key Constraints

- Also referred to as Entity Constraints.
- Key constraints force that :
  - in a relation with a key attribute, no two tuples can have identical values for key attributes.
  - a key attribute can not have NULL values.

Domain Constraints

- Attributes have specific values in real-world scenario.
  - E.g. age can only be a positive integer
  - E.g. age cannot be less than zero, telephone numbers cannot contain a digit outside 0-9

Referential integrity Constraints

- Referential integrity constraints work on the concept of Foreign Keys.
  - A foreign key is a key attribute of a relation that can be referred in other relation.
- Referential integrity constraint states that if a relation refers to a key attribute of a different or same relation, then that key element must exist.

References

- https://www.guru99.com/dbms-keys.html
- https://www.numpyninja.com/post/types-of-keys-in-dbms